# Fanyi Meng

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## Education

**The Chinese University of Hong Kong (Shenzhen) (CUHKSZ)** Sep. 2023 – Sep. 2025

*M.Phil.* in Computer and Information Engineering (CIE)

- GPA: **3.46 / 4**

**Southern University of Science and Technology (SUSTech)** Sep. 2019 – Jun. 2023

*B.S.* in Electrical and Electronic Engineering (EEE), Communication Engineering

- GPA: **3.47 / 4** (85 / 100)

## Experience

**Southern University of Science and Technology (SUSTech)** Oct. 2025 – Present

*Research Assistant* Supervisor: Tianyue Zheng

- Developed high-fidelity wireless channel fields utilizing **Fourier Neural Operators (FNO)** based on 3D geometric models.
- Collected multi-modal sensing data utilizing an **ASUS Router** for **Channel State Information (CSI)** extraction and a **Livox Mid-40** LiDAR.

**Shenzhen Research Institute of Big Data** Mar. 2023 – Jul. 2025

*Research Assistant* Supervisor: Guangxu Zhu

**DIIS** Laboratory Internal Project

- Extracted **CSI** from Wi-Fi signals on Intel 5300 NIC, ESP32, and Router RT-AC86U.
- Built a Wi-Fi Sensing System for data collection, model training, and function visualization.
- Used **BERT**-based neural network to address **packet loss** challenges in Wi-Fi Sensing tasks.
- Employed three Intel 5300 NICs for sub-2-second latency in real-time human **tracking and localization**.

**Guangdong Basic and Applied Basic Research Foundation Project**

- Calibrated **Sionna** Ray-tracing (RT) models with real-world beam-wise RSRP.
- Simulated **system-level** network performance metrics (e.g., Spatial Efficiency, Localization Error) incorporating advanced 6G technologies (RIS, ISAC).
- Optimized network performance using **gradient-based** methods, leveraging the chain rule for differentiation within Sionna's Ray-tracing and Neural Network components.
- Leveraging **Taichi** to enable differentiable, flexible, and high-performance ray-tracing simulations.

**Southern University of Science and Technology (SUSTech)** Oct. 2020 – Jun. 2023

*Undergrad Research Assistant* Supervisor: Rui Wang

Project with **HONOR**

- Used **MATLAB's WLAN Toolbox** for simulation of the Distributed Coordination Function (DCF) within the Wi-Fi MAC layer.
- Used **NS-3** to analyze the Enhanced Distributed Channel Access (EDCA) mechanism in 802.11ac, focusing on throughput and latency.
- Refined EDCA parameters to improve overall **QoS** performance in Channel Allocation for Wi-Fi networks.

## Publications

⋆ **Fanyi Meng**, Tianyue Zheng. "Euler-Fi: An Eulerian Mixture-of-Experts for Rapid Wireless Channel Field Modeling."
*Submitted to IEEE Transactions on Mobile Computing*, 2026.
**Abbreviation:** IEEE TMC **| Under Review | CCF Category:** CCF-A

- ⋆ **Fanyi Meng**, Xinhao Li, Dongzhu Liu, Guangxu Zhu. "Environment Calibrated and Fully Differentiable Wireless Network Simulator for 6G."
  *Submitted to IEEE Communications Magazine*, 2026.
  **Abbreviation:** IEEE Commun. Mag. **| Under Review | CCF Category:** CCF-B
- ⋆ Zijian Zhao, **Fanyi Meng**, Z. Lyu, Hang Li, Xiaoyang Li, Guangxu Zhu. "CSI-BERT2: A BERT-inspired Framework for Efficient CSI Prediction and Classification in Wireless Communication and Sensing."
  *IEEE Transactions on Mobile Computing*, 2026.
  **Abbreviation:** IEEE TMC **| Pages:** 7241-7257 **| CCF Category:** CCF A
- ⋆ Zijian Zhao, Tingwei Chen, **Fanyi Meng**, Hang Li, Xiaoyang Li, Guangxu Zhu. "Finding the Missing Data: A BERT-inspired Approach Against Package Loss in Wireless Sensing."
  *IEEE INFOCOM DeepWireless Workshop*, 2024.
  **Abbreviation:** INFOCOM Wkshps **| Pages:** 1-6 **| CCF Category:**Workshop of CCF A Conference
- ⋆ Yihang Jiang, **Fanyi Meng**, Xinhao Li, Xiaoyang Li, Guangxu Zhu, Kaifeng Han, Qingjiang Shi. "Coverage Analysis for Air-Ground Integrated-Sensing-and-Communication Networks."
  *International Conference on Ubiquitous Communication*, 2024.
  **Abbreviation:** Ucom **| Pages:** 455-459
- ⋆ Xinhao Li, Yihang Jiang, **Fanyi Meng**, Xiaoyang Li, Kaifeng Han, Guangxu Zhu. "Integrated Sensing and Communications Signal Prediction with Multi-Beam Statistical Channel."
  *International Symposium on Antennas and Propagation*, 2024.
  **Abbreviation:** ISAP **| Pages:** 1-2

## Skills

- Programming Languages: Python > MATLAB > C++ > LabVIEW = Java
- Platforms: Ubuntu 20.04, Debian 10, USRP X410
- Tools: Taichi, Nvidia Sionna, CSI-Tool, TensorFlow, NS-3, PyTorch, AirSim, ROS

## Honors and Awards

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| *3rd Prize*, the First Wi-Fi Sensing Contest | Dec. 2023 |
| Advanced to the semifinals in International Algorithm Competition of Pazhou Lab | Nov. 2023 |

## Miscellaneous

- Website: `http://fanyimeng0.github.io`
- GitHub: `https://github.com/fanyimeng0`
- Languages: English - TOEFL 91, Mandarin - Native speaker